Dhruv Meshram

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ACADEMIC DETAILS

Degree	Specialization	Institute	Year	CPI/%
B.Tech.	*Computer Science and Engineering*	Indian Institute of Technology Gandhinagar	2024–Present	7.73
Class XII	*Physics, Chemistry, Maths*	St. Paul Junior College, Nagpur	2022–2024	87.17
Class X		Narayana Vidyalayam, Nagpur	2021–2022	95.4

PROJECTS

- **CPM Planner — Project Management & Optimization** [May 2026 – June 2026]
  GitHub Code | *Live Application*
  - Engineered **sub-millisecond path resolution** across complex project networks containing over **50k+ activity nodes** by developing a compiled **C++ cpm_engine** optimized with Kahn's algorithm and dynamic programming to execute depth-first topological sorting, cycle detection, and scheduling pass operations
  - Optimized **dependency query latency by 45%** during database benchmarks of highly nested graphs by implementing **clustered indexing** and recursive common table expressions (CTEs) to resolve deeply chained scheduling constraints and eliminate query planning bottlenecks
  - Streamlined **graph payload serialization by 38%** during engine-to-API data exchange by architecting a custom inter-process communication (IPC) channel that streams binary data packets to bypass heavy JSON serialization overhead and minimize end-to-end execution latency
  - Accelerated **scheduling simulation throughput by 55%** under complex resource constraints by designing a heuristic resource leveling algorithm that dynamically resolves allocation bottlenecks across parallel critical paths, tasks, and dependency chains
- **Freshwash DBMS — High-Performance Sharded Database** [January 2026 – April 2026]
  GitHub Code
  - Reduced **disk I/O overhead by 35%** across database transactions by designing a custom **C++ B+ Tree engine** integrated with sequential **Write-Ahead Logging (WAL)** for page-level crash recovery
  - Sustained **92% throughput retention** under multi-threaded contention by implementing **latch crabbing protocols** and node-level reader-writer locks across a concurrent **B+ Tree structure**
  - Streamlined **node-routing latency by 40%** across decentralized storage clusters by implementing a **consistent hashing algorithm** to balance partition distribution across active nodes
  - Accelerated **primary-key lookup speed by 65%** across complex relational schema queries by implementing a multi-table **index-nested join** framework leveraging the underlying **B+ Tree**
- **VidyutAI Dashboard — Machine Learning & MLOps** [February 2026 – April 2026]
  (Prof. Pallavi Bharadwaj, IIT Gandhinagar) | *GitHub Code*
  - Benchmarked a **94.2% MAPE** load forecasting system by evaluating sequential **LSTM** neural networks against gradient-boosted trees (**XGBoost**) to analyze structural architectural trade-offs
  - Structured **500k+ grid records** into an optimized feature engineering pipeline, boosting peak-demand capture by **22%** via high-dimensional lag variables, rolling windows, and seasonal Fourier terms
  - Engineered a **35% compute reduction** by benchmarking tabular **XGBoost** against recurrent **LSTM** models to analyze sequential depth and inference latency trade-offs on time-series test sets
  - Optimized an **18% error reduction** under anomalous peak-load grid conditions by integrating weather-interaction features and implementing walk-forward validation to handle extreme seasonal volatility

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C, SQL, HTML/CSS, Verilog
- **Machine Learning & Frameworks:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, ONNX, Next.js, Flask, Node.js, Zustand
- **Databases & Infrastructure Tools:** PostgreSQL, MongoDB, Prisma ORM, Socket.io, Git, Docker, Linux

RELEVANT COURSEWORK

- Data Structures and Algorithms, Software Tools and Techniques for AI, Database Management Systems (DBMS)

EXTRA-CURRICULAR ACTIVITIES

- **BAJA SAE Team Member:** Fabricated off-road vehicle components for national competitive motorsport environments
- **Recreational Interests:** Badminton, motorsports, Formula 1, and automotive engineering

POSITIONS OF RESPONSIBILITY

- **Team Member, Moksha Motorsports, BAJA SAE Team, IIT Gandhinagar** [Mar 2025 – Present]
  - Contributed to the brakes department for the eBAJA competition, collaborating within a multidisciplinary team to design, simulate, and optimize off-road vehicle prototypes
  - Managed and streamlined team-wide technical documentation, ensuring comprehensive records of design reviews, engineering specifications, and operational updates